

فیزیولوژی

1- What is the cause of "physiological shunt" occurrence?

Decreasing the cardiac output to half



Decreasing of ventilation/perfusion ratio (VA/Q) to zero

Mixing of pulmonary vein blood with bronchial vein blood

Increasing of VA/Q ratio more than normal amount

2- Which zone(s) is (are) seen in a healthy subject during exercise?

Zones II and III



Zone-III

Zone-I and II

Zone-II

3- Which one of the following is TRUE about "the duration of expiration"?

Increases by strong activation of pneumotaxic center



Decreases when respiratory rate increases

Is less than duration of inspiration during normal and quite respiration

Is affected by ventral respiratory group (VRG) during normal respiration

4- In which one the following, the diffusion rate of gases is decreased between two sides of respiratory membrane?

Increasing the gas pressure difference between the two sides of the membrane

Increasing the surface area



Removal of part of the lungs

Decreasing the thickness of the membrane

5- In the lungs, which one of the following is NOT a barrier to gas exchange between capillary and the alveolar space?

A fluid layer of water and surfactants covering the luminal surface of alveolar cells

Alveolar epithelial cells



Smooth muscle cells surrounding the endothelial cells

Basement membranes of endothelial and alveolar cells

6- Which one of the following is TRUE about bronchial circulation?

- ✓ Supplies blood flow to conducting zone
- Provides pulmonary capillary perfusion for gas exchange
- Has vasoconstriction in response to hypoxia
- Is a part of the the low-pressure circulation

7- Which of the following lung volume or capacity is more than the others?

- Expiratory reserve volume
- ✓ Inspiratory reserve volume
- Residual volume
- Functional residual capacity

8- What is the the initial pressure difference (mmHg) that causes O₂ to diffuse into the pulmonary capillary?

- 21
- ✓ 64
- 17
- 11

9- What is the effect of an increase in the CO₂ of blood?

- ✓ Decreases the affinity of oxyhemoglobin for oxygen
- Decreases dissociation of O₂ from oxyhemoglobin
- Shifts the oxyhemoglobin dissociation curve to the left
- Increases the amount of oxyhemoglobin

10- Which one of the following center is thought to control of the respiratory rate?

- Ventral respiratory group of medullary neurons
- ✓ Pneumotaxic center
- Dorsal respiratory group of medullary neurons
- Apneustic center

11- What is "the maximum amount of air that can be expired after a maximum inspiration"?

- ☒ Vital capacity
- Total lung volume
- Expiratory reserve volume
- Tidal volume

12- If a patient had a progressive lung disease that required an ever increasing pressure to fill the same volume of lung, how would the lung's compliance be affected?

- It would increase it.
- It would stay the same
- ☒ It would decrease it.
- These variables do not affect lung compliance.

13- Given your knowledge regarding the response of the pulmonary vasculature to oxygen content, where is the highest amount of blood flow through a normally oxygenated lung in a normal weight adult male that is standing upright?

- To the midzone of the lung
- ☒ To the bottom of the lung
- To the top of the lung
- There is no gradient across the lung

14- In response to hypoxia, which neurotransmitter is NOT released by "Glomus cells" in the carotid bodies?

- Acetylcholine
- Dopamine
- ☒ Histamine
- ATP

15- What is the chloride shift in RBC of venous blood?

- HCO₃⁻ goes into the cell and chloride ion goes out of the cell
- chloride ion goes out of the cell and H⁺ goes into the cell
- ☒ chloride ion goes into the cell and HCO₃⁻ goes out of the cell
- H⁺ goes into the cell and chloride ion goes out of the cell

مدت آزمون: 34 بر اساس دفترچه کاربر respiratory آزمون

16- If a patient's blood carries 10 grams of hemoglobin per deciliter, what is the maximum O2 carrying capacity of blood (milliliters per deciliter) in this person?

19.8

14.4

☒ 13.4

20.1

17- Which one of the following gas has more "diffusing capacity" than others in resting?

Carbon monoxide

☒ Carbon dioxide

Oxygen

Nitrogen

18- Which one of the following leads to shift to the left of O2 dissociation curve?

Increased metabolism

☒ Decreased H⁺ concentrationIncreased PCO₂

Decreased pH

19- Which one of the following has afferent nerve fibers in the vagus nerve?

Chemosensitive area in medulla

☒ Aortic body

Hypoxia- induced vasoconstriction

Carotid body

20- Which one of the following list includes only structures that compose the anatomic dead-space?

Respiratory bronchioles, alveolar ducts, alveoli

Pharynx, bronchi, alveoli, larynx

☒ Trachea, bronchi, terminal bronchioles

Respiratory bronchioles, alveoli, trachea

21- What is the minute respiratory volume (L/min); with a tidal volume of 0.8 L and respiratory rate of 15 breaths/min?

8.5

4.2



12

6

22- Which one of the following occurs during normal inspiration?



Diaphragm contracts, pleural and alveolar pressures decrease

Diaphragm relaxes, external intercostals contract, pleural pressure increases

External and internal intercostals contract, pleural and alveolar pressure increase

Diaphragm contracts, pleural pressure increases, alveolar pressure decreases

بافت

23- In which of the following is observed Stratified Squamous Tissue?

Pharynx

Olfactory Zone

Respiratory Zone



Vestibule Zone

24- What Is The Name Cell Secreted Surfactant In Bronchioles?

Macrophage

Simpl Cells

Pneumocyte III



Celara

25- Which of the following is observed in structure Histology In Respiratory Zone?

Dense Connective Tissue + Erectile Tissue

Stratified Squamous Epithelial Tissue

Dense Connective Tissue



Goblet Cell

26- In which of the following is observed Elastic Fiber In Histology Stractuer ?

- ✓ Larynex
- Respiratory Zone
- Vestibule Zone
- Alveolar Duct

27- Which part of the lung tissue has Nomocyte II?

- ✓ Alveolar
- Terminal Bronchiole
- Bronchus
- Bronchiole

آناتومی

28- Which bone is involved in both the roof and the medial surface of the nasal cavity?

- Ethmoid
- Inferior concha nasalis
- ✓ Palatine
- Sphenoid

29- Which of the following muscles is involved in pulling thyroid cartilage downward and forward ?

- ...
- Posterior crico- arytenoid
- ✓ Cricothyroid
- Oblique arytenoid
- Transverse arytenoid

30- Which of the following is the highest structure in the right lung umbilicus?

- Pulmonary artery
- ✓ Superior upper lobe bronchus
- Pulmonary vein
- Main bronchus

31- The medial surface of the right lung is adjacent to all the following structures except

Vena cava superior

Azygos vein



Arch of aorta

Vena cava inferior

32- Which of the following paranasal sinuses open to the apex of bulla ethmoidal ?



Middl ethmoidal sinus

Maxillary

Anterior ethmoidal sinus

Frontal

33- What is the name of the upper and free side of the quadrangular membrane?



Ary -epiglottic fold

Vesibular ligament

Median thyrohyoid ligament

Vocal ligament

34- Which of the following nerve pass through incisive canal ?

Greater palatine

Posterior ethmoidal



Nasopalatine

Anterior ethmoidal

35- All of the following bones participates in the formation of the lateral wall of the nose except?



Vomer

Lacrimal

Palatine

Maxilla

36- All of the following structures are seen in inlet of larynx except?

Ary-epiglottic fold



Vestibular fold

Epiglot cartilage

Arythnoid cartilage

37- Which of the following arteries passes through the foramen in roof of nasal cavity?

Greater palatine artery



Anterior ethmoidal artery

Superior labial artery

Sphenopalatin artery

38- All of the following muscles are paralyzed by cutting the recurrent laryngeal nerve except?

Lat Crico- arytenoid



Cricothyroid

Thyroepiglottic

Posterior crico- arytenoid

جنین شناسی

39- In which period of lungs development alveolar duct appear



Canalicular phase

Pseudoglandular phase

Alveolar phase

Saccular phase

40- In which time of development the second epithelial alveolar cells appear?

3th month



6th month

5th month

4th month